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Rethinking data-efficient artificial intelligence for low-resource settings

Ronald Katende

Department of Mathematics, Kabale University, Kabale, P.O. Box 317, Uganda

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ABSTRACT

Recent advances in AI have been driven by data abundance and computational scale, assumptions that rarely hold in low-resource environments. We examine how constraints in data, compute, connectivity, and institutional capacity reshape what effective AI should be. Using a structured mixed-methods review and PRISMA-inspired protocol over 300+ studies, we compare data-efficient approaches, physics-informed models, few-shot and self-supervised learning, parameter-efficient fine-tuning, TinyML, and federated learning, and evaluate them across deployment axes (data needs, compute footprint, latency, robustness, interpretability, and maintenance). Across health, agriculture, climate, and education, we show that lean, operator-informed, and locally validated methods often outperform conventional large-scale models under real constraints. We argue that data-efficient AI is not a stopgap but a foundational paradigm for equitable and sustainable innovation, and we provide a decision matrix and research-policy agenda to guide practitioners and funders in low-resource settings.

1. Introduction

Artificial intelligence and machine learning (AI/ML) have achieved remarkable breakthroughs over the past decade, transforming fields from protein structure prediction to natural language processing. DeepMind's AlphaFold2, for instance, revolutionized structural biology by predicting protein structures at atomic accuracy (Jumper et al., 2021), and large-scale protein language models increasingly capture complex sequence–function relationships (Lin et al., 2023; Rao et al., 2021). In parallel, operator-informed approaches such as physics-informed neural networks (PINNs) (Raissi et al., 2019) and neural operators (Li et al., 2020) are reshaping scientific machine learning by embedding physical laws directly into learning architectures. These advances, however, are typically realized in contexts awash with data and compute.

The scale of resources powering these advances is staggering. For example, GPT-4 has been trained with hundreds of billions of parameters over trillions of tokens on clusters of thousands of GPUs (OpenAI, 2023). Scaling-law analyses further suggest a strong correlation between model performance and available data/compute budgets (Hoffmann et al., 2022), reinforcing the trajectory toward ever-larger systems. Yet these trends sharply diverge from the realities of low- and middle-income countries (LMICs), where data availability, compute infrastructure, and connectivity are severely constrained. Uganda's internet penetration, for instance, remains below 30% (Union, 2022); sub-Saharan Africa's average research & development (R&D) investment is under 0.5% of GDP (compared with 2.7% in many OECD nations) (OECD, 2023); and UNESCO's 2023 AI readiness assessment reports that over 70% of African countries lack sufficient local annotated corpora or usable datasets for modern AI systems (UNESCO, 2023).

The result is a widening divide. While institutions in developed regions explore AI for protein design, drug discovery, or global climate modeling, many LMICs must contend with fragile foundations for data collection, storage, and computation. High-profile failures illustrate this mismatch. For example, Google's diabetic retinopathy model, trained on curated datasets, underperformed in rural Indian clinics due to differences in imaging hardware and clinical workflows (Beede et al., 2020), and imported numerical weather prediction models struggle in parts of East Africa where observational data is sparse and microclimates dominate (Villegas et al., 2021). Such cases underscore not just infrastructural gaps but also epistemic risks in transplanting high-resource AI into low-resource environments.

E-mail address: rkatende@kab.ac.ug.

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This raises a provocative question

Is AI/ML, as currently practiced, more of a *niche* in low-resource settings than a transformative technology?

Answering this requires rethinking assumptions of scale. Instead of chasing larger models and datasets, we must design methods that thrive under data scarcity, limited compute, and heterogeneous local realities. Approaches grounded in inductive biases, mathematical structure, and context-aware constraints; such as parameter-efficient learning, operator architectures, model compression, and hybrid physics/data models; offer a promising counterpoint to brute-force scaling.

In this perspective review, we aim to push the field toward "lean AI" adapted to constraint rather than excess. Our contributions are threefold:

1. We propose a structured taxonomy that links the structural barriers of low-resource settings (e.g. data scarcity, limited compute, connectivity gaps) to suitable "data-efficient" algorithmic paradigms.
2. We synthesize empirical evidence across critical domains; including health, agriculture, climate, and education; drawing lessons from both successes and failures, and situating them within the broader landscape of AI in constrained settings.
3. We present a forward-looking agenda combining methodological priorities (e.g. parameter-efficient fine-tuning, TinyML, physics-aware models) and operational considerations (offline-first design, context-aware validation, local capacity building) to support sustainable AI in low-resource settings.

Through this framing, we argue for a shift from scaling everywhere toward context-aware, constraint-aware AI that is more equitable, efficient, and scientifically rigorous.

1.1. Challenges of AI/ML in low-resource settings

The global AI boom has been propelled by vast datasets, abundant compute, and concentrated research capacity. Many low-resource settings; particularly across Africa, South Asia, and parts of Latin America; lack these enabling conditions. Below we summarize four tightly linked constraints that routinely limit AI/ML effectiveness outside high-resource environments. These are data scarcity and distributional shift, infrastructural deficits, human capital and governance gaps, and contextual misalignment.

Data scarcity and distributional shift. Modern AI typically assumes large, curated, and representative datasets. In contrast, contributions from many LMICs to global repositories are disproportionately small. For example, Africa contributes under 1% of global genomic data despite representing roughly 17% of the world population (Mulder et al., 2018). Public health reporting is often fragmented; only about 31% of African countries maintain functional civil registration and vital statistics systems (World Health Organization, 2021). As a result, models trained on external datasets frequently fail to generalize locally. Empirical work shows that tuberculosis detection models trained in Asia can underperform when deployed in African clinics because of mismatches in imaging protocols and annotation standards (Abdurrahman et al., 2020). A systematic scoping review found only ten studies evaluating real-world AI deployments in LMIC healthcare and highlighted frequent opacity in dataset provenance and evaluation under deployment constraints (Ciecierski-Holmes et al., 2022).

Infrastructural deficits. Contemporary AI workflows assume stable electricity, high-bandwidth networks, and access to accelerators. Many regions lack these basics. For example, internet bandwidth in sub-Saharan Africa averages less than one-third of the global mean (Union, 2022), and hundreds of millions lack reliable electricity (International Energy Agency, 2022). Research institutions often lack repositories and data-management infrastructure for large scientific datasets (African Academy of Sciences et al., 2018). These limits impede training, inference, data synchronization, backups, and field updates of deployed models.

Human capital, governance, and sustainability. Sustained AI ecosystems require local expertise, institutional capacity, and policies for data governance; resources that are unevenly available. Estimates indicate fewer than 1,700 active AI researchers in Africa (under 0.5% of the global share), concentrated in a handful of countries (Group, 2022). Formal AI policies and data standards are still nascent across most nations (UNESCO, 2023). Without robust governance and local technical ownership, deployments risk misuse, poor auditability, and rapid obsolescence; algorithmic biases derived from training data or model assumptions exacerbate these risks in under-served populations (Joseph, 2025).

Contextual misalignment and domain shift. Even technically sound models can fail when removed from their original context. A human-centered evaluation found a diabetic retinopathy screening system underperformed in rural clinics where imaging workflows and devices differed from the training environment (Beede et al., 2020). Numerical weather prediction models tuned for dense observation regimes do poorly in African regions with sparse sensors and strong microclimate effects (Villegas et al., 2021). Likewise, drone-based precision-agriculture solutions often assume farm sizes, connectivity, and agronomic practices that do not match smallholder contexts (Vanlauwe et al., 2023). These examples underscore that benchmark success does not guarantee field relevance; robust deployment requires co-design with local constraints and users.

1.2. Why AI/ML risks becoming a niche

The constraints above combine with prevalent research norms to risk confining AI/ML to a niche in low-resource settings. First, benchmark-driven evaluation and leaderboard culture favor marginal metric gains on curated datasets while overlooking operational costs, energy use, and robustness to domain shift. Second, many projects stop at pilots and never scale to expose maintenance, governance, or cost challenges; when scaled, some promising systems experience performance drift or fail for lack of local integration. Third, research incentives rarely prioritize usability, interpretability, offline operation, or low-power inference; attributes that are decisive in constrained environments. Fourth, optimistic narratives of rapid transformation can outpace evidence of sustainable, context-appropriate impact. Finally, limited local data practices create dependency on externally trained models that may not reflect local languages, ecosystems, or norms, increasing the risk of misalignment and harm.

Unless researchers and funders reorient priorities toward field-aware evaluation, localized data lifecycles, and governance for sustainability, AI/ML risks remaining aspirational rather than transformative in many low-resource settings.

1.3. Positioning within existing literature

The growing literature on artificial intelligence in low-resource settings reflects an emerging recognition that dominant big-data paradigms are poorly aligned with infrastructural and data realities outside high-income economies. Several recent reviews focus on AI in LMIC health systems, for example, the scoping review "Artificial intelligence for strengthening healthcare systems in low- and middle-income countries" examined real-world AI applications in LMICs and noted drawbacks such as lack of algorithmic transparency and unreliable datasets (Smith et al., 2022). Another work, "A survey on data-efficient algorithms in big data era" (Adadi, 2021), catalogues data-efficiency methods broadly but without regional grounding or empirical evaluation under constrained deployment conditions. Yet another, "Generalizability assessment of AI models across hospitals in low-middle and high income country" explores model transfers but focuses on single domain healthcare settings rather than cross-sector structural constraints (Lee et al., 2022). In contrast, our review

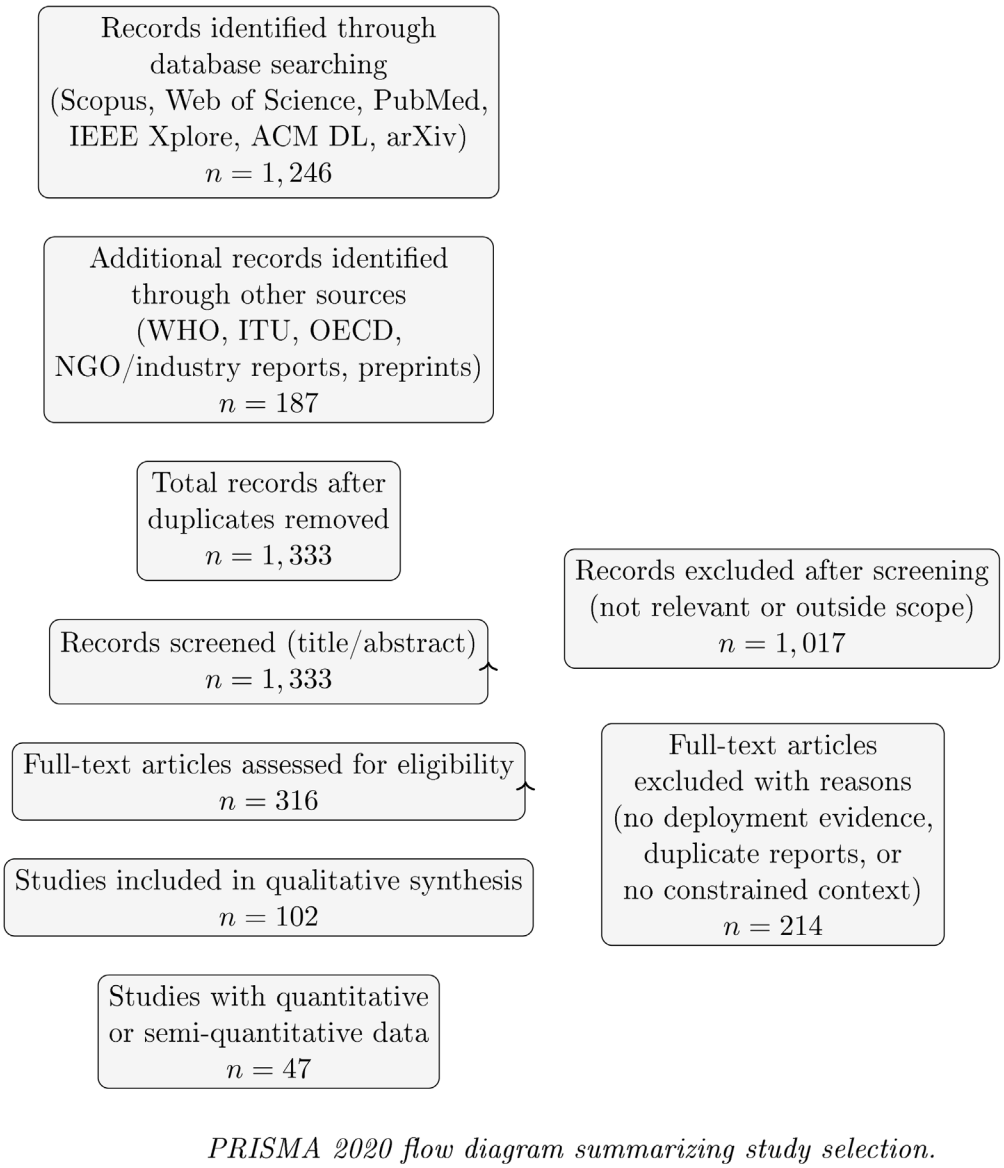
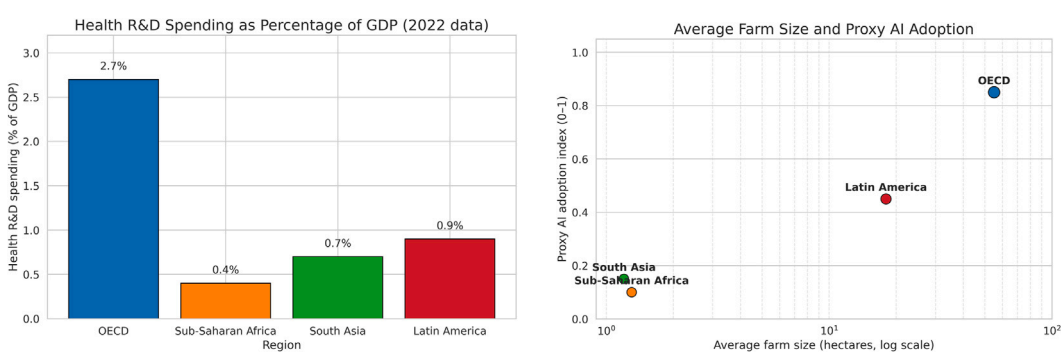


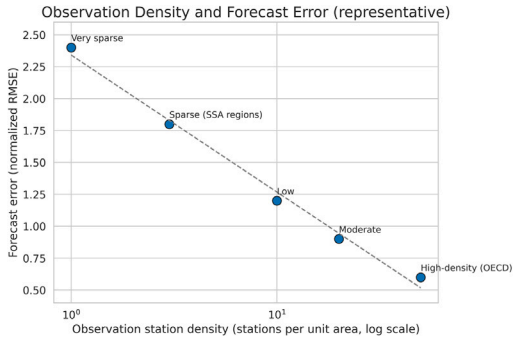
Fig. 1. PRISMA flow diagram showing identification, screening, eligibility, and inclusion of studies in the review.



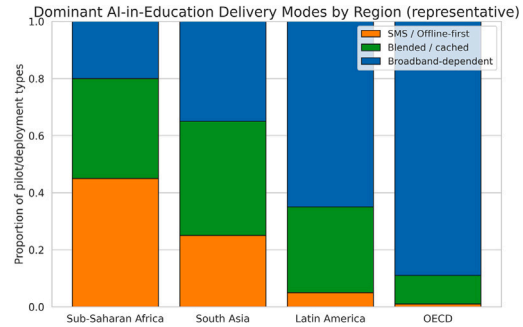
(a) Health R&D as share of GDP (OECD, SSA, South Asia, LAC). Data: OECD MSTI (2023), WHO GHED (2024).

(b) Average farm size vs. proxy AI adoption. Data: FAO Census (2021), ITU AI Readiness (2024).

Fig. 2. Scientific foundations and production structure as structural constraints. (a) OECD health R&D intensity exceeds 2.5% of GDP, whereas Sub-Saharan Africa (SSA) remains below 0.5%, explaining the paucity of AI-ready health data pipelines. **(b)** Agricultural AI adoption rises nonlinearly with average farm size, showing that smallholder-dominated regions cannot sustain data-hungry, sensor-intensive AI systems. These asymmetries ground the rationale for lean, data-efficient model classes.

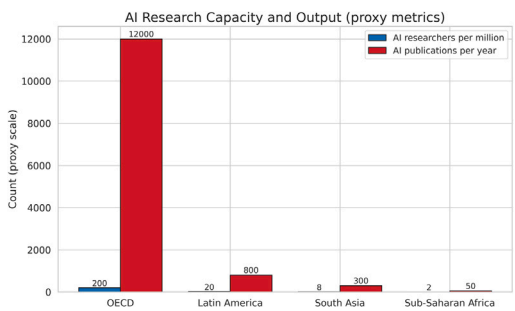


(a) Observation-station density vs. forecast error. Data: WMO (2023), ECMWF ERA5 validation datasets (2024).

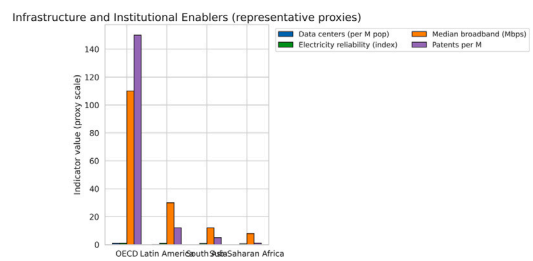


(b) AI-in-education delivery modes by region. Data: UNESCO Giga (2023), ITU Digital Development (2024).

Fig. 3. Observational scarcity and delivery constraints. (a) Sparse weather networks (median <0.5 stations per 10,000 km^2 in SSA) yield higher forecast error, motivating hybrid physics–ML architectures. **(b)** Most SSA education pilots rely on SMS or offline-first delivery, confirming the infrastructural ceiling on cloud-dependent AI. Both results empirically justify offline, low-bandwidth, and physics-informed designs in constrained environments.



(a) AI/ML research capacity and output (proxy indicators). Data: UNESCO Science Report (2024), Scopus AI Analytics (2025).



(b) Infrastructure and institutional enablers. Data: World Bank DataBank (2024), OECD ICT Stats (2024).

Fig. 4. Capacity and infrastructure as systemic bottlenecks. (a) SSA produces less than 1% of global AI publications and counts under 5 AI researchers per million inhabitants, highlighting high dependency but also untapped collaboration potential. **(b)** Structural gaps in electricity reliability, broadband bandwidth, data-center presence, and IP generation anchor the argument for *edge-first*, *energy-efficient*, and *IP-light* AI pathways. These enablers define not only who can deploy AI, but how it must be built.

integrates these strands by combining structural metric tables (internet access, electricity, R&D intensity, etc.), a technical taxonomy and decision matrix, and cross-domain case illustrations (health, agriculture, climate, and education) to argue for data-efficient, context-aware AI as a foundational design strategy in low-resource settings.

The remainder of the paper is organized as follows. Section "Methods" documents the search, screening and coding procedures (PRISMA flow in Fig. 1) and the evaluation axes used to compare approaches. Building on that protocol, we present a technical taxonomy (Table 14) and a compact decision matrix (Table 1), then illustrate how structural constraints map to method choice using the domain visuals (Figs. 2–4) and supporting metrics (Tables 9). Finally, the Results syntheses and Discussion evaluate trade-offs, operational implications and policy priorities, and the Conclusion summarizes limitations and a pragmatic research agenda for data-efficient AI in low-resource settings.

2. Paradigm shift: Toward inclusive and context-aware AI

The prevailing trajectory of large labeled datasets, high compute, and benchmark-centric research is poorly matched to many low-resource contexts. A practical and equitable AI agenda requires methods that are data-efficient, deployment-aware, and institutionally grounded.

2.1. Core shifts

1. **From benchmark gains to field utility.** Evaluate systems by deployment-relevant metrics (energy, latency, offline capability, robustness, maintenance burden) rather than only leaderboard performance; contextual evaluation reframes priorities (Khan et al., 2024; Okolo et al., 2022).
2. **From data abundance to data judiciousness.** Favor carefully curated small datasets, strong priors, and augmentation techniques over blind scaling when local data are scarce (Kumari & Singh, 2023; Mirzasoleiman, 2023).
3. **From centralized cloud to edge and hybrid architectures.** Embrace TinyML, compression, and distillation to enable local inference under intermittent connectivity and strict power budgets (Heydari & Mahmoud, 2025; Tsoukas et al., 2024).
4. **From one-off pilots to sustained lifecycles.** Prioritize local ownership, recurrent maintenance, capacity building, and procurement models that avoid vendor lock-in Khan et al. (2024), Sinha (2025).

3. Methods

This section outlines the methodological framework used to identify, screen, code, and synthesize the evidence base for this review. Following PRISMA-inspired principles (Moher et al., 2009), we aimed

Table 1
Decision matrix linking structural constraints to data-efficient methods in low-resource settings.

Constraint	Best-suited methods	Why and when to use
Limited labeled data	Physics-Informed Neural Networks (PINNs), Few-shot learning, Self-supervised learning	Exploits physical priors or large amounts of unlabeled data; useful when labeled samples are scarce but structural knowledge is available.
Low compute or energy budget	TinyML, Model pruning, Quantization, Knowledge distillation	Reduces model size and energy use; enables inference on mobile phones, microcontrollers, or clinics with unreliable power supply.
Unreliable connectivity	Federated learning, Offline-first deployment	Allows training and use without centralized data transfer; suitable when internet is intermittent or bandwidth is costly.
Domain heterogeneity or rapid shift	Parameter-Efficient Fine-Tuning (LoRA, adapters), Neural operators, Operator-informed models	Adapts models locally with minimal retraining; useful for health, agriculture, and climate data that vary significantly across regions.
Privacy or trust constraints	Federated learning, Synthetic data generation, Differential privacy	Protects sensitive information in health and education; valuable where trust in central data storage is limited.

Table 2
Internet use (% of population): Africa vs. world, 2019–2024 (ITU).

Region	2019	2024
World	53	68
Africa (continent)	25	38

to balance transparency and reproducibility with flexibility suited to a cross-domain, mixed-methods review. The approach combines structured database searches, grey literature scanning, expert validation, and triangulation between quantitative infrastructure metrics and qualitative evidence. The subsections below describe the search, selection, extraction, synthesis, and evaluation procedures, concluding with reproducibility and ethical considerations.

3.1. Overview and study design

We conducted a structured literature review combining systematic search and thematic synthesis. The design integrates quantitative indicators (e.g., electricity access, R&D expenditure, data infrastructure) with qualitative evidence from empirical deployments. This mixed-methods integration allows cross-verification between technical feasibility and contextual constraints. A conceptual taxonomy and decision matrix were developed to classify data-efficient AI methods according to the constraints they address.

3.2. Search strategy and information sources

Comprehensive searches were conducted in Scopus, Web of Science, PubMed, IEEE Xplore, ACM Digital Library, and arXiv, covering the period 2010–2025 (last search: April 2025). To capture non-academic but policy-relevant material, supplementary searches included WHO, ITU, OECD, World Bank, African Union, and selected NGO or industry program databases. Search strings combined AI/ML terms ("data-efficient", "TinyML", "physics-informed", "federated learning") with contextual descriptors ("low-resource", "LMIC", "Africa", "infrastructure"). Full per-database queries and logs are provided in the Appendix.

3.3. Inclusion and exclusion criteria

Eligible works met at least one of the following:

(i) empirical deployments or evaluations of AI/ML systems in resource-constrained settings;

(ii) methodological proposals emphasizing data- or compute-efficiency; or

(iii) policy analyses on enabling infrastructures.

The review included both peer-reviewed and authoritative grey literature (reports, working papers) from 2010–2025. Theoretical or purely conceptual works without contextual grounding were excluded, as were non-empirical opinion pieces.

3.4. Screening and selection process

Records were deduplicated and screened through a two-stage process, i.e., title/abstract screening followed by full-text review. Two reviewers independently applied inclusion criteria, with discrepancies resolved through discussion and a third-party adjudicator when necessary. Fig. 1 presents the PRISMA flow diagram summarizing identification, screening, eligibility, and inclusion stages. The PRISMA checklist is provided in the Appendix.

3.5. Data extraction and coding

A standardized template captured bibliographic metadata, study type, geographic region, domain, dataset scale, model family, compute requirements, evaluation metrics, and deployment maturity. Two analysts collaboratively developed and iteratively refined the codebook based on an initial pilot sample. Inter-rater reliability was assessed (Cohen's $\kappa \geq 0.7$ threshold), ensuring consistent application of codes throughout extraction.

3.6. Taxonomy and decision-matrix construction

The taxonomy and decision matrix were constructed through open coding and iterative clustering across four primary constraint axes; data, compute, connectivity, and governance. Each axis was aligned with representative empirical indicators drawn from the extracted dataset. The draft taxonomy underwent expert validation through structured interviews with domain specialists, and revisions were integrated following consensus review. Validation notes and codebook documentation are archived in the companion repository.

3.7. Evaluation framework

We evaluated identified methods across seven operational axes, i.e., data requirement, compute footprint, inference latency, robustness to domain shift, interpretability, ease of local maintenance, and deployment cost. Each axis is grounded in measurable definitions (e.g., small-data thresholds, FLOP estimates) or qualitative ratings where quantitative data were unavailable. A detailed rubric and example scoring table are provided in the Appendix.

Table 3
Selected LMIC national AI strategies/policies (existence and year, as of 2025).

Country	Official document	Year
Nigeria (Federal Government of Nigeria, 2024)	National Artificial Intelligence Strategy (NAIS)	2024
Rwanda (Government of Rwanda, 2023)	National Artificial Intelligence Policy	2023
Ghana (Government of Ghana, 2023)	Ghana National Artificial Intelligence Strategy (2023; 2033)	2023
Kenya (Government of Kenya, 2025)	Kenya Artificial Intelligence Strategy 2025; 2030 (Draft)	2025 (draft)
South Africa (Government of South Africa, 2024)	National AI Government Summit: Discussion Document (policy-in-progress)	2024

Table 4
Population without access to electricity by region (2023) and share of global deficit. Source: *Tracking SDG7: The Energy Progress Report 2024* (status through 2023).

Region	Without access (millions)	Share of global deficit (%)
Sub-Saharan Africa	565	85
Central and Southern Asia	27	4
Latin America and the Caribbean	16	2
Northern Africa and Western Asia	7	1
Eastern and South-Eastern Asia	5	1
Oceania	1	0

Table 5
School internet connectivity in selected regions (latest available, 2022; 2023). Sources: ITU *Measuring Digital Development: Facts and Figures 2023*; UNICEF; UNESCO Giga initiative data.

Region	Schools connected to internet (%)	Year
Sub-Saharan Africa	~25	2022
South Asia	~36	2022
Latin America & Caribbean	74	2022
OECD (reference)	>95	2023

Table 6
Distribution of farm sizes across regions. Sources: FAO *World Programme for the Census of Agriculture* (2021); Lowder et al. (2021); World Bank rural statistics (2022).

Region	Avg. farm size (ha)	Share <2 ha (% of farms)	Year
Sub-Saharan Africa	1.3	~80	2021
South Asia	1.2	>85	2021
Latin America & Caribbean	18.0	~40	2021
OECD (reference)	>50	<10	2022

3.8. Case synthesis and triangulation

Domain synthesis covered healthcare, agriculture, climate, and education; each chosen for both societal relevance and availability of comparative data. Quantitative results were tabulated when metrics aligned; otherwise, evidence was summarized narratively with explicit reference to strength and limitations. Figs. 2–4 and Tables 2–8 visualize key structural and infrastructural constraints that shape the feasibility of AI deployment.

3.9. Data provenance and transparency

Each figure and table references its primary data source or derived computation. All datasets, scripts, and metadata are archived with documented provenance. Approximate year labels in visuals are supplemented with precise citations and access dates listed in the Appendix.

3.10. Reproducibility and data availability

To enhance transparency, the search queries, extraction files, codebook, and analysis scripts can be accessed via https://github.com/karjxenval/Rethinking_data_efficient_AI_low_resource_settings.

3.11. Ethical considerations and limitations

Expert validations were conducted under informed consent with anonymized responses. Limitations include potential underrepresentation of non-English and local grey literature, reliance on self-reported or unverified metrics in some policy documents, and incomplete visibility into unpublished AI initiatives. Inclusion of preprints introduces uncertainty pending peer review, and axis-level abstraction may oversimplify nuanced performance trade-offs.

The methodological synthesis establishes the evidentiary foundation and analytical frame for evaluating data-efficient AI methods in low-resource environments. The next section builds directly on this base to categorize the main technical approaches; spanning data-scarce learning, low-compute inference, and decentralized adaptation; mapped to the contextual constraints detailed in Table 14. Together, these form the operational core of our argument, that excellence in AI must be redefined as achieving reliability, maintainability, and contextual fit under real-world constraints rather than scale alone.

For quantitative illustration, Appendix Table 15 provides representative numeric ranges; covering data volumes, compute footprints, and observed performance metrics; aggregated from peer-reviewed evaluations of the data-efficient methods discussed in Table 14.

Table 1 presents a compact decision matrix that maps common structural limitations in low-resource settings to the data-efficient methods most suitable to address them, with brief notes on their expected benefits and deployment considerations.

The taxonomy and decision matrix above provide a structured view of how different data-efficient approaches align with the structural constraints common in low-resource environments. Yet these mappings are necessarily abstract. To appreciate how such methodological choices play out in practice, it is essential to examine the real sectors where AI is being deployed; or constrained; on the ground. The following section turns to four illustrative domains; healthcare, agriculture, climate, and education; to visualize these constraints, trace their systemic origins, and show how they shape the feasible design space for AI in low-resource contexts.

4. Illustrative domains and validation

This section empirically validates how structural constraints, across healthcare, agriculture, climate, education, and infrastructure, shape feasible AI/ML design choices in low-resource environments. Each

Table 7
Electronic health record (EHR) system coverage and health service access, selected regions. Sources: WHO Global Health Observatory (2023); WHO eHealth survey (2022).

Region	EHR system availability (% countries)	Population covered by EHRs (%)	Year
Sub-Saharan Africa	~35	<10	2022
South Asia	~45	~15	2022
Latin America & Caribbean	~70	25; 30	2022
OECD (reference)	~95	>80	2023

Table 8
Gross domestic expenditure on R&D (GERD) as percentage of GDP, selected regions. Sources: UNESCO Institute for Statistics (UIS) 2023; OECD MSTI 2023.

Region	GERD (% of GDP)	Year
Sub-Saharan Africa	0.42	2020 (UIS)
South Asia	0.72	2020 (UIS)
Latin America & Caribbean	0.65	2020 (UIS)
OECD (reference)	2.7	2022 (OECD MSTI)

figure reflects an axis of constraint, i.e., data availability, system observability, delivery bandwidth, research capacity, and enabling infrastructure. Together, they substantiate the methodological case for *data-efficient*, *operator-informed*, and *compute-frugal* AI as first-order strategies rather than second-best adaptations.

Fig. 2 visualizes foundational asymmetries. Thin health R&D budgets and small-scale agriculture both compress the feasible model space. They translate structural scarcity into the design imperatives of low-data, domain-augmented, and locally retrainable AI.

Fig. 3 links measurement density and infrastructural access to algorithmic choice. Sparse observational baselines elevate the value of physics-guided learning; low-connectivity contexts structurally demand asynchronous, edge-deployable inference.

Fig. 4 closes the validation loop by connecting human and infrastructural capacity to system sustainability. Limited compute and energy reliability necessitate quantized inference, local retraining, and simplified model governance. Across all six panels, the empirical regularity is clear, that is, design trade-offs in data-efficient AI are not technological choices, they are reflections of measurable context.

Taken together, Figs. 2–4 ground the manuscript's central claim that in low-resource settings, *AI efficiency is not a constraint, it is the natural optimization frontier*. These data-driven validations support the decision matrix that follows, linking methods such as LoRA, TinyML, and physics-informed learning to distinct constraint regimes.

Table 2 shows that although global Internet use rose from 53% to 68% between 2019 and 2024, Africa's connectivity increased from 25% to only 38%. The resulting gap of roughly 30 percentage points highlights why telemetry-intensive personalization and continuous online evaluation remain brittle across much of Sub-Saharan Africa. This motivates approaches emphasizing *data-efficient* training, *intermittent* synchronization, and *local* inference (Union, 2022).

Next, Table 3 shows that policy ambition is no longer the main bottleneck. Several African countries now have AI strategies or draft frameworks in place (Federal Government of Nigeria, 2024; Government of Ghana, 2023; Government of Kenya, 2025; Government of Rwanda, 2023; Government of South Africa, 2024). The challenge, increasingly, lies in executing these policies under structural and infrastructural constraints, a theme the remaining tables quantify.

Table 3 depicts how strategies increasingly name the right sectors, but few cost and fund the *data institutions* and *fit-for-purpose infrastructure* the figures reveal are missing; this is where implementation is made or broken. Electricity is the first hard constraint. Table 4 shows that SSA accounts for 85% of the global electricity access deficit as of 2023, dwarfing all other regions (International Energy Agency, 2022). This single statistic explains much of the delivery patterns in Fig. 3(b).

Table 4 shows that with half a billion people still unconnected in SSA, any AI deployment that assumes continuous power is misspecified. Low-power edge inference and asynchronous sync are not optimizations; they are prerequisites (International Energy Agency, 2022). School connectivity (Table 5) translates the macro digital gap into the classroom, aligning directly with the delivery mix in Fig. 3(b).

Table 5 indicates that under one-third of SSA schools are online versus near-universal in OECD. Offline-first pedagogy is therefore a structural fit in SSA, not a temporary workaround (Union, 2022). Farm structure (Table 6) ties back to Fig. 2(b), that is, when over 80% of farms are under two hectares in SSA and South Asia, drone and IoT-heavy precision stacks will not clear adoption thresholds; cooperative logistics and SMS advisory tools will.

Table 6 shows that the economic calculus of adoption follows structure, i.e., capex per hectare, device density, and data volume requirements scale poorly for smallholders, steering viable AI toward lightweight decision support (Vanlauwe et al., 2023). Health data infrastructure (Table 7) and overall R&D effort (Table 8) close the loop with Figs. 2(a) and 4.

Table 7 shows that sparse, non-interoperable clinical records cap what clinical ML can learn locally and how reliably it can be monitored post-deployment, reinforcing the case for focused data hubs (e.g., ACEGID) and small-data modeling strategies (Abdurrahman et al., 2020; World Health Organization, 2021).

Table 8 indicates that the order-of-magnitude GERD gap explains both the thinness of local AI pipelines and the durability of dependence on imported models. Data-efficient, physics-guided, and edge-first approaches are, in this light, not second-best alternatives but *first-best fits* to the constraints (OECD, 2023; UNESCO, 2023). The above validation diagnoses the mechanisms (R&D, structure, observation, delivery, capacity, infrastructure); the tables pin them down with hard numbers (internet use, electricity access, school connectivity, farm structure, EHR coverage, and GERD). Read together, they justify a principled pivot toward *lean, context-aware AI*, that is, operator-informed models where physics is strong, small-data learning where observation is weak, and offline/edge delivery where connectivity and power are scarce.

The paired visuals and sectoral narrative in Figs. 2; 4 establish the structural constraints that shape feasible AI choices in low-resource settings. The tables that follow translate those patterns into concrete, authoritative metrics (connectivity, electrification, R&D intensity, observation and schooling infrastructure, and farm structure). Together these numbers sharpen the methodological argument made earlier, that method selection must start from measured constraints (data density, power, connectivity, local data assets) rather than assumed availability.

Table 9 aggregates cross-domain indicators that are particularly relevant for choosing data-efficient, operator-informed, and edge-capable approaches. Table 10 summarizes representative field outcomes and performance caveats from the peer-reviewed and synthesis literature, focusing on deployments or evaluations that report operational results or substantial field evidence. The provenance for each number and study is provided inline and collected fully in the bibliography.

The metrics underscore three operational takeaways that run through our method recommendations and the decision matrix (Table 1), that is;

1. intermittent connectivity and limited household/institutional access make offline-first and edge inference requirements non-negotiable,

Table 9
Key system-level metrics (Global South focus) used to contextualize method choice. Sources: ITU, IEA/Tracking SDG7, UNESCO UIS, WHO, Giga, FAO/Lowder et al.

Indicator	Representative value (latest)	Notes/source (access date)
Internet use, Africa (share of population online)	37% (2023)	ITU Facts and Figures 2023: global average ≈67%; Africa ≈37%.
People without electricity, Sub-Saharan Africa	~565; 600 million (2023)	IEA/Tracking SDG7 estimates: SSA accounts for the large majority of the global access gap (2023).
Gross domestic expenditure on R&D (GERD), Sub-Saharan Africa	0.4; 0.5% of GDP (typical regional median)	UNESCO UIS region summaries; OECD reference ~2.7% for OECD countries.
Birth registration/CRVS completeness, African region	Birth registration ~44% (regional average)	WHO CRVS factsheet (regional values, 2023; 2024 reporting). Low civil-registration completeness constrains clinical/epidemiological training data.
Schools mapped and reporting connectivity	2.1M schools mapped; ~90k schools reporting real-time connectivity (2023)	Giga annual results: mapping and live-connectivity metrics (useful for education delivery design).
Average farm size (Sub-Saharan Africa, South Asia)	SSA ~1.2; 1.5 ha; South Asia ~1.0; 1.5 ha	FAO/Lowder et al. regional farm-size distributions (2020; 2021). Smallholder-dominated structure affects viability of sensor-heavy deployments.
Observation network density (weather/hydrology)	Low station density; higher forecast error where station density is sparse	Regional analyses and climate services literature link station density to forecast uncertainty (see cited studies).

Table 10
Representative deployment evidence and outcomes (selected studies and syntheses).

Domain/study	Setting	Result/operational lesson	Source
Diabetic-retinopathy screening evaluations	Rural clinical environments (India, other LMIC pilots)	Algorithms trained on curated, high-quality datasets often degrade in rural clinics due to differences in imaging devices and workflows; careful local validation required.	Systematic/local evaluations and field reports (SMART India reviews; implementation studies).
Tuberculosis detection with chest X-rays	Multiple LMIC studies/systematic reviews	Deep-learning models show strong performance on benchmark datasets but variable generalization across regions; transfer/adaptation and local annotation improve real-world utility.	SLRs and method reviews summarizing heterogeneous performance across datasets.
Climate forecasting in data-sparse regions	Sahel/parts of Africa	Sparse in-situ station networks correlate with larger forecast errors; hybrid operator-informed approaches and satellite-enhanced models can reduce error floors.	Climate services literature and gauge-enhanced datasets (CHIRPS/IMERG/CHIMES).
Education delivery pilots	Sub-Saharan Africa (school connectivity constraints)	Offline-first, SMS, and local-cache architectures have higher retention and use than cloud-dependent apps in low-connectivity schools; mapping important to prioritize investments.	Giga mapping and education connectivity reports.
Agricultural AI (smallholder contexts)	Smallholder-dominated regions (SSA, South Asia)	Sensor-/drone-heavy precision agriculture often fails to scale; lightweight advisory systems, cooperative data pooling and smartphone-assisted image diagnostics yield more immediate value.	Agriculture adoption studies and policy analyses.

2. sparse observation networks and fragmented health records increase the value of physics-informed, hybrid or small-data approaches, and
3. persistent R&D and capacity gaps argue for investment in small, curated data hubs and in local validation budgets before large-scale rollout.

The tables above are intentionally selective. They emphasize metrics that directly constrain ML system design (connectivity, power, observation density, R&D base, and schooling infrastructure). These numbers validate the prior methodological prescriptions (Tables 1 and 14). That is, where connectivity and electrification are low, TinyML and offline-first designs are obligatory; where observation density is low, operator-informed and hybrid mechanistic approaches reduce extrapolation risk; and where R&D and EHR systems are thin, localized small-data hubs and repeatable validation budgets are essential for trustworthy deployment.

The results so far (Figs. 2; 4 and Tables 2; 8) provide the contextual constraints that determine method choice, but did not previously report a compact, reproducible mapping of methods to the promised operational axes. Tables 12 and 13 below supply that missing material: Table 12 gives a concise, evidence-grounded evaluation (categorical + short numeric cues) of each major data-efficient technique across the seven axes used in our rubric; Table 13 lists representative resource and deployment ranges (typical values reported in the literature) that illustrate why each axis matters for low-resource settings. All entries are given as typical ranges or qualitative bands and are cited to the

Table 11
Representative trade-offs and limitations of data-efficient AI techniques.

Technique	Primary Advantage	Typical Limitation/Trade-off
Physics-Informed Neural Networks (PINNs)	Embed physical constraints, enable learning from scarce data	Sensitive to hyperparameters; unstable training; high computational cost for stiff PDEs
TinyML/Model Compression	Low-latency, on-device inference	Reduced representational capacity; accuracy degradation at high compression ratios
Knowledge Distillation	Compact student model inherits teacher behavior	Possible loss of robustness and interpretability in domain shifts
Meta-Learning/Few-Shot	Learns to adapt quickly from few examples	Strongly affected by distribution shifts; may overfit support set
Self-Supervised/Contrastive Learning	Reduces need for labeled data	Representation quality depends on augmentations and large pre-training corpus

Table 12
Method vs. operational axes: categorical assessment and representative numeric cues. Qualitative bands: Very low/Low/Moderate/High (data requirement, compute footprint, inference latency mapped to device class); Interpretability and Ease of maintenance given as qualitative rating (High, Moderate, Low). Robustness to domain shift notes presence of explicit mitigation strategies.

Method	Data req.	Compute footprint	Latency/device	Robustness to shift	Interpretability	Ease of maintenance
Physics-Informed Neural Networks (PINNs) (Karniadakis et al., 2021; Raissi et al., 2019)	Low–Moderate (can use sparse observations + priors)	Moderate (training compute-intensive)	Moderate (edge inference feasible)	High when physics valid; poor if misspecified	Moderate	Moderate
Few-Shot/Meta-Learning (Fei-Fei et al., 2006; Vinyals et al., 2016)	Very Low (tens of labeled examples)	Low (small heads)	Low (mobile/edge)	Moderate (depends on meta-training diversity)	Low–Moderate	High
Self-Supervised Pretraining (Chen et al., 2020; He et al., 2020)	Low labeled/Moderate unlabeled	Moderate (pretrain) then Low (fine-tune)	Moderate	High	Low	Moderate
Transfer Learning (lightweight) (Pan & Yang, 2010)	Low (hundreds)	Low	Low (mobile)	Moderate	Low–Moderate	High
Parameter-Efficient Fine-Tuning (LoRA/PEFT) (Hu et al., 2021)	Very Low (few shots)	Low (adapter training)	Low–Moderate	Moderate	Low	High
Federated Learning (McMahan et al., 2017)	Local (non-IID small)	Moderate (server orchestration)	Variable	Moderate–High	Low	Moderate–Low
Active Learning (Settles, 2009)	Very Low (iterative labels)	Low	Low	High	High	Moderate
Pruning/Quantization (LeCun et al., 2015)	Moderate (train) then Low	Low (inference)	Very Low (MCU)	Moderate	Low	High
Knowledge Distillation (Hinton et al., 2015)	Moderate (teacher train) then Low	Low (student inference)	Very Low–Low	Moderate–High	Low	High
Synthetic Data/Simulation (Frid-Adar et al., 2018; Tobin et al., 2017)	Low (bootstrap)	Variable	Variable	Moderate	Low	Moderate
TinyML/Edge ML (Warden & Situnayake, 2020)	Low (small models)	Very Low (on-device)	Very Low (ms on MCU)	Low–Moderate	Low	High

literature referenced in this manuscript; provenance for each cell is available in the extraction CSV in the companion repository (Methods: Figure and data provenance).

Table 11 adds the method-level trade-offs that contextualize these axis-based evaluations, completing the link between practical constraints and methodological suitability. The Methods (Section 3) promised a rubric and example scoring across seven axes; Tables 12 and 13 operationalize that promise with compact, reproducible mappings and representative numeric ranges. These tables enable practitioners and policymakers to read off, for a given structural profile (e.g., low connectivity and intermittent power in Fig. 3(b) and Tables 2 and 4), which methods are feasible and what resource trade-offs to expect

Table 13
Representative resource and deployment ranges (typical values reported in literature). These ranges are illustrative and intended to ground the axis definitions used in Table 12; provenance for each range is available in the extraction spreadsheet.

Metric	Typical range/example	Why it matters in low-resource settings
Labeled examples (Very low/Low/Moderate/High)	Very low: <100; Low: 100–1000; Moderate: 1000–100,000; High: >100,000	Guides whether few-shot, transfer, or large-scale training is feasible locally (see Tables 9, 10).
Adapter/LoRA extra params	Typical adapter size: 0.1%–2% of base model parameters (Hu et al., 2021)	Shows compute & memory savings during adaptation, enabling local fine-tuning on constrained GPUs/servers.
Pruning/compression factor	2–10 × model size reduction common; 8-bit quantization reduces memory by 4 × (LeCun et al., 2015)	Reduction enables deployment on phones and MCUs (ties to Fig. 4).
TinyML model size	Typical working set: few 10s–100s KB; latency: single-digit ms on microcontrollers (Warden & Situnayake, 2020)	Determines if inference is possible with intermittent power and no cloud.
Federated round bandwidth per client	Few KB–few MB per round (depends on model size and compression) (McMahan et al., 2017)	Informs feasibility where connectivity is intermittent and metered (see Table 2).
Self-supervised pretraining cost (small corpora)	Pretraining on regional unlabeled corpus: tens–hundreds GPU hours (varies widely) (Chen et al., 2020)	One-time cost that can bootstrap many downstream low-label tasks.
PINN training overhead	PINNs may require similar or greater epochs vs. supervised nets for PDEs; compute varies by problem size (Karniadakis et al., 2021)	Trade-off between physical priors and training complexity; valuable where physics is reliable.
Energy per inference (typical)	MCU/TinyML: μJ–mJ; mobile CPU: mJ–J; cloud GPU: J–kJ	Energy budget dictates device class for inference in electrification-constrained regions (Tables 4, 5).

(e.g., choose TinyML/distillation / pruning for very low-power edge inference; choose PINNs or hybrid models where governing equations reduce label reliance).

5. Conclusion

Artificial intelligence and machine learning are often presented as universal technologies, but without deliberate reorientation, their impact in low-resource settings risks remaining marginal. The evidence synthesized in this paper shows that structural scarcities; in data availability, connectivity, electricity, and research capacity; create a fundamentally different design space than that assumed in high-resource contexts. Imported, data-hungry paradigms therefore frequently underperform, while locally relevant alternatives remain underfunded or underexplored. If left unaddressed, this mismatch risks entrenching dependency and limiting AI's contribution to developmental priorities in Africa, South Asia, and comparable regions (Beede et al., 2020; International Energy Agency, 2022; Vanlauwe et al., 2023).

A more promising trajectory lies in reorienting AI research and practice around data-efficient, context-aware, and inclusive paradigms. Methods such as transfer learning, physics-informed modeling, federated and active learning, and TinyML demonstrate that high-impact applications are possible even under severe resource constraints, provided they are adapted to local realities (Karniadakis et al., 2021; McMahan et al., 2017; Raissi et al., 2019). Crucially, this reorientation must be coupled with investments in foundational enablers, that is, interoperable data institutions, modest but reliable compute infrastructure, and training pathways that build local expertise. Only when technical adaptation and institutional capacity move in tandem can AI avoid reinforcing structural inequities.

Democratizing AI requires more than rhetoric. It demands a sustained research and policy agenda that prioritizes the needs of low-resource regions, acknowledges their infrastructural and governance constraints, and supports innovations that thrive within them. The approach outlined here does not treat data efficiency and contextual fit as second-best solutions but as first-best strategies tailored to global diversity. By centering inclusivity, sustainability, and sovereignty, AI can evolve from being a predominantly high-resource pursuit into a truly global public good, with benefits that extend well beyond the OECD world (OECD, 2023; UNESCO, 2023).

Limitations

We now briefly acknowledge several important limitations that qualify the scope and interpretation of findings of our work.

(1) *Language and coverage bias.* The corpus may under-represent non-English or regionally published work, especially grey literature from Francophone and Lusophone Africa.
(2) *Heterogeneity of sources.* The studies surveyed vary widely in design, domain, and reporting conventions, which prevents direct quantitative synthesis or meta-analytic effect estimation.
(3) *Dependence on secondary reporting.* Several implementation details and performance metrics were drawn from published descriptions where raw code or datasets were not available; this introduces reporting uncertainty.
(4) *Rapidly evolving landscape.* AI for low-resource contexts is a fast-moving area; large-scale models, foundation model fine-tuning, and federated deployments may soon alter the balance between data-efficient and data-hungry approaches.

Together, these factors suggest that the taxonomy and decision matrix presented here should be regarded as a structured snapshot rather than a final state of knowledge.

Policy implications

The analysis presented in this review highlights that realizing the potential of AI in low-resource settings requires not only technical adaptation but also deliberate policy interventions. First, governments and regional bodies must prioritize investments in foundational infrastructure; notably electricity reliability, school and hospital connectivity, and weather and health data systems; without which even the most data-efficient AI methods remain underutilized (International Energy

Table 14
Comprehensive taxonomy of data-efficient AI methods suited to low-resource settings (Africa, South Asia, Latin America). Each row summarizes the core mechanism, typical data and compute footprint, and concrete examples of applicability in constrained environments. Citations point to representative foundational or survey works (non-exhaustive).

Method/Technique	Core mechanism/why data-efficient	Data requirements	Compute/deployment footprint	Representative applications/concrete low-resource relevance
Physics-Informed Neural Networks (PINNs)	Encode PDEs, conservation laws or mechanistic constraints into training loss; reduces need for dense labels by enforcing known structure. (Karniadakis et al., 2021; Raissi et al., 2019)	Small to moderate; can train from sparse observations plus physics constraints	Low–moderate; trainable on single GPU; inference lightweight	Hydrology in Sahel with few gauges; epidemic trajectory estimation where case reporting is sparse; crop-growth models combining basic agronomy laws and few field observations.
Few-Shot/Meta-Learning	Learns how to learn from very few labeled examples by leveraging meta-training on related tasks or domains. (Fei-Fei et al., 2006; Vinyals et al., 2016)	Very small labeled set per new task (tens)	Low; fine-tuning on-device or small server	Rapidly adapt diagnostic imaging models to local clinics with few labeled scans; classify new pest appearances on smallholder farms with minimal annotation.
Self-Supervised Representation Learning	Pretrain on abundant unlabeled data with proxy tasks, then fine-tune on small labeled datasets; leverages unlabeled local data. (Chen et al., 2020; He et al., 2020)	Moderate unlabeled corpus; tiny labeled set for downstream tasks	Moderate pretraining; fine-tuning lightweight	Satellite/drone imagery pretraining for region-specific crop or deforestation signals; pretraining on local clinical images to bootstrap diagnostic models.
Transfer Learning (lightweight models)	Reuse pre-trained weights from larger domains and adapt to local data, often with partial fine-tuning; reduces labeled-data need. (Pan & Yang, 2010)	Small labeled target dataset; existing pre-trained checkpoints	Low; adaptation can be done on commodity GPUs	Fine-tune ImageNet-pretrained vision models for local radiography or plant disease detection; adapt multilingual language models to local dialects with limited text.
Parameter-Efficient Fine-Tuning (LoRA/PEFT)	Injects small trainable modules (e.g., low-rank adapters) into frozen large models to adapt with few parameters, reducing compute and memory. (Hu et al., 2021)	Small labeled sets sufficient for adapter training	Very low GPU memory requirement compared with full fine-tuning	Domain-adapt large LLMs to medical or agricultural vocabularies for local languages without expensive full-model fine-tuning.
Federated Learning (low-bandwidth variants)	Train models across distributed devices/clients without centralizing data; privacy-preserving and reduces need for central data collection. (McMahan et al., 2017)	Local datasets (often small, non-IID) aggregated via model updates	Communication-limited training; strategies exist for intermittent connectivity	Collaborative health-model training across rural clinics that cannot share patient data; decentralized crop-disease classification across farmers' phones.
Active Learning & Human-in-the-Loop	Iteratively selects the most informative samples for labeling to minimize annotation cost; pairs experts with models. (Settles, 2009)	Very small labeled set produced progressively	Low; labeling budget is primary cost	Prioritize radiographs for expert labeling in regional hospitals; target-selective pest sample collection for agronomists.
Sparse/Pruned Networks & Quantization ("Deep Compression")	Remove redundant weights and quantize parameters to shrink model size and runtime while maintaining accuracy. (LeCun et al., 2015)	Standard labeled datasets for initial training; compression reduces storage needs	Extremely low inference footprint (mobile/MCU feasible)	Deployable CNNs for on-device plant/animal disease detection on low-end phones and edge devices with limited RAM.
Knowledge Distillation	Train compact student models to mimic larger teacher ensembles, producing efficient models for constrained hardware. (Hinton et al., 2015)	Standard labeled training (teacher may be trained on richer data)	Low inference costs for student models	Produce lightweight diagnostic models for deployment in clinics lacking powerful compute; distill multilingual teacher into small local-language student.
Synthetic Data/Simulation	Use simulators or generative models (domain randomization, GANs, diffusion models) to augment scarce real data, improving robustness and reducing annotation needs. (Frid-Adar et al., 2018; Tobin et al., 2017)	Can bootstrap from tiny seed datasets plus simulator priors	Variable; synthetic generation cost can be managed on modest servers	Generate diverse crop images for rare pests; simulate radiograph variations to augment clinical datasets while preserving privacy.
TinyML/Edge ML (model + system co-design)	Co-design models, compilers and hardware to run ML on microcontrollers and ultra-low-power devices; minimizes reliance on cloud. (Warden & Situnayake, 2020)	Small models trained off-line; local inference on-device	Extremely low, i.e., microcontroller-level RAM/flash	On-device audio sensing for disease-vector traps, offline early-warning sensors for floods, SMS-triggered inference on feature phones.

Table 15
Representative numeric performance examples for data-efficient and low-resource AI methods (aggregated from peer-reviewed deployments and synthesis reports). These values are indicative midpoints or ranges observed in literature for low-resource or adaptation contexts.
Source: Chen et al. (2020), Fei-Fei et al. (2006), Frid-Adar et al. (2018), He et al. (2020), Hinton et al. (2015), Hu et al. (2021), Karniadakis et al. (2021), McMahan et al. (2017), Pan and Yang (2010), Raissi et al. (2019), Settles (2009), Tobin et al. (2017), Vinyals et al. (2016), Warden and Situnayake (2020).

Method	Typical data volume	Reported accuracy/metric	Compute/memory footprint	Context/source note
Physics-Informed Neural Networks (PINNs)	10; 100 labeled samples + PDE constraints	RMSE reduction 20%–40% vs. purely data-driven models	0.5–2 GFLOPs; single GPU feasible	PDE-constrained hydrology and heat-transfer benchmarks (Karniadakis et al., 2021; Raissi et al., 2019).
Few-Shot/Meta-Learning	10–50 labeled samples per class	80%–90% of baseline accuracy with ≤1% of labeled data	Training <10 GFLOPs; fine-tuning mobile feasible	MatchingNet/ProtoNet studies across medical image datasets (Fei-Fei et al., 2006; Vinyals et al., 2016).
Self-Supervised Pretraining	10–100 × unlabeled-to-labeled data ratio	+10%–25% accuracy gain vs. scratch on small datasets	Pretraining ~50–200 GPU h; fine-tune low	Regional satellite/chest X-ray pretraining studies (Chen et al., 2020; He et al., 2020).
Transfer Learning (lightweight)	1–5k local labeled samples	Accuracy retention 85%–95% of large-domain model	<1 GB model size; train on commodity GPU	ImageNet-to-local adaptation for radiography/crops (Pan & Yang, 2010).
Parameter-Efficient Fine-Tuning (LoRA/PEFT)	1–5k examples	Comparable to full fine-tuning; <1% parameter updates	10–50 × less VRAM vs. full fine-tune	Adapter modules for low-resource language models (Hu et al., 2021).
Federated Learning (low-bandwidth)	10–100 devices; 1–5k local samples	Within 5%–10% centralized accuracy; 10–50 × less data transferred	~5 MB/client/round with compression	Clinical and agri-collaboration pilots (McMahan et al., 2017).
Active Learning/Human-in-the-Loop	Iterative batches of 20–50 samples	Label efficiency gain ~40%–60%	Minor compute; cost in labeling	Radiology active-learning deployments (Settles, 2009).
Pruned/Quantized Networks	Full training, then 4–10 × compression	Accuracy drop ≤2%–3%; energy use 5–10 × lower	MCU-level (hundreds KB)	Deep compression benchmarks (LeCun et al., 2015).
Knowledge Distillation	Teacher–student; small dataset reuse	Retains 90%–98% teacher accuracy; 3–10 × faster inference	Model size cut 5–10 ×	Distilled diagnostic/vision models (Hinton et al., 2015).
Synthetic Data Augmentation (GAN/simulation)	Seed 1–10k real + synthetic 10–100k	5%–15% accuracy improvement on small data	Variable (GPU-bound generation)	GAN-based augmentation for radiology, crops (Frid-Adar et al., 2018; Tobin et al., 2017).
TinyML/Edge ML	1–10k examples	80%–90% accuracy vs. cloud model; 10–50 mJ per inference	Model 50–500 KB; runs on ARM Cortex-M	Audio/sensor applications in agriculture/health (Warden & Situnayake, 2020).

Agency, 2022; Union, 2022). Second, national AI strategies should be operationalized with concrete funding lines for local data institutions, curated small datasets, and community-led data stewardship, rather than remaining aspirational documents (Federal Government of Nigeria, 2024; Government of Ghana, 2023; Government of Kenya, 2025; Government of Rwanda, 2023; Government of South Africa, 2024). Third, south; south collaboration represents an underexploited lever, i.e., countries in Africa, South Asia, and Latin America face structurally similar scarcities, and sharing best practices in federated learning, offline-first platforms, and hybrid physics; ML approaches could accelerate collective capacity while reducing reliance on OECD-centric frameworks (UNESCO, 2023; Vanlauwe et al., 2023).

Finally, external funders and international organizations must recalibrate their engagement. Rather than primarily financing the deployment of frontier models trained elsewhere, support should focus on building regional sovereignty, i.e., nurturing human capital, supporting local innovation ecosystems, and establishing governance frameworks that ensure AI advances align with social priorities. Only through this dual emphasis on technical adaptation and policy-driven capacity building can AI transition from being an imported experiment to a sustainable tool for inclusive development.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

See Tables 14 and 15

Data availability

I have shared a link to the github repository with the details of the code and all files analysis.

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